

# Fundamentals of Printing Science

Articulate the fundamental knowledge of color, human visual perception, color theories, and illumination concepts for accurate color

OFFICIAL SOURCE DECLARATION

## How this lesson was prepared

The supplied official SITTTTR syllabus PDF for Course 2101 is the authoritative source for course information, objectives, course outcomes, CO-PO mapping, modules, topics, activities, assessment structure and references. Explanations, study prompts and Malayalam support are educational elaboration and are not presented as additional official syllabus text.

**Source file:** 2101.pdf from the uploaded official SITTTTR syllabus archive. **Course code and title:** preserved from the source.

COURSE DASHBOARD

## Official course information

Course information

| Item         | Official information             |
|--------------|----------------------------------|
| Programme    | Diploma Programme                |
| Course code  | 2101                             |
| Course title | Fundamentals of Printing Science |
| Semester     | II                               |

| Item                 | Official information                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|----------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Course category      | As specified in the official PDF                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| Periods per week     | As specified in the official PDF                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| Periods per semester | As specified in the official PDF                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| Credits              | As specified in the official PDF                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| Prerequisite         | Topic Course code Course Title Semester Electro magnetic spectrum, Newton's color wheel, Engineering Physics for Mechanical, Structural and 2002A I/II Light, Viscosity, Industrial Applications Course Outcome Blooms Taxonomy CO Course Outcome Level Explain the principles of color and human visual perception and apply color harmony, color mixing, and CO1 Understand illumination concepts for accurate and consistent color reproduction in printing. Articulate digital materials for printing by applying basic knowledge of image quality, color reproduction, CO2 Understand and image input processes used in print production. Explain how images are reproduced with tonal variation and how print quality is evaluated and controlled CO3 Understand using basic measurement concepts. Explain how the characteristics of printing substrates and inks influence printability, machine performance, CO4 Understand and final print quality CO-PO mapping COURSE ARTICULATION MATRIX PROGRAM OUTCOMES |
| Assessment           | Use the official assessment section reproduced below                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |

## Modules

4 official module sections detected.

## Topic entries

33 source-aligned topic entries retained.

## Study mode

Theory and application emphasis.

### STUDY METHOD

## How to use this lesson

### First pass

Read the official source declaration, dashboard and module transcript. Mark unfamiliar terms without changing the official terminology.

## Second pass

Use each topic card to build a definition, principle, steps or components, application and exam answer. Attempt the Q&A before opening the answers.

## Practical evidence

Where the course is laboratory or workshop based, maintain an observation notebook with procedure, instruments, readings, safety points and conclusion.

## Revision pass

Return to the source excerpt, coverage table, activities and model paper. The generated paper is practice material, not an official examination paper.

### LEARNING OUTCOMES

## Objectives, outcomes and mapping

### Course objectives

- Articulate the fundamental knowledge of color, human visual perception, color theories, and illumination concepts for accurate color
- understanding and reproduction in printing.
- effectively during print production.
- operation and acceptable print quality.

### Course outcomes

| CO  | Official outcome      | Study level           |
|-----|-----------------------|-----------------------|
| CO1 | 3 2 - 1 1 1 - - - - 1 | See official syllabus |
| CO2 | 3 2 1 1 2 1 - - - - 1 | See official syllabus |

| CO  | Official outcome                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Study level           |
|-----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------|
| CO3 | 3 3 1 2 1 1 - - - 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | See official syllabus |
| CO4 | 3 2 1 1 1 2 1 1 - - 1 Total 12 9 3 5 5 5 1 1 - - 4 Correlation Level 3 2.25 1 1.25 1.25 1 1 - - 1 Legends: 1 - Low; 2 - Medium; 3 - High; No Correlation - “-” Teaching and Assessment Scheme Diploma Curriculum Revision 2026 2101 Page #1 Examination Scheme Teaching Scheme/Week Self-learning Total Credits Theory Practical (SS)/Week Hours/Semester C Total L T P S CIE ESE Total CIE ESE Total 4 0 0 0 6 60 4 40 60 100 0 0 0 100 L: Lecture (One unit is of one-hour duration with one credit), T: Tutorial (One unit is of one-hour duration with one credit), P: Practical (One unit is of one-hour duration with 0.5 credit), S: Project (One unit is of one-hour duration with 0.5 credit) SS: Self-Learning, CIE: Continuous Internal Evaluation, ESE: End Semester Examination Syllabus - Major Topics Instructional Module Title Major Topics Hours L T Color: Definition. Attributes of color - Hue, Saturation, Lightness, Alliance of EM Spectrum, Wavelength of Light and Color, Human Color perception - Elements of eye, Rod and LMS Cone cell vision. Standard illuminant - Definition, Types of illuminants, Color temperature. Spectral | See official syllabus |

#### CO-PO mapping evidence

| Row | Extracted official mapping |
|-----|----------------------------|
| CO1 | CO1 3 2 - 1 1 1 - - - 1    |
| CO2 | CO2 3 2 1 1 2 1 - - - 1    |
| CO3 | CO3 3 3 1 2 1 1 - - - 1    |
| CO4 | CO4 3 2 1 1 1 2 1 1 - - 1  |

#### COVERAGE AUDIT

## Complete syllabus coverage

Every detected official module is represented below. Each module contains topic cards and an expandable official syllabus transcript to preserve source terminology.

#### Module coverage

| Module   | Official heading                            | Topic entries | Hours        |
|----------|---------------------------------------------|---------------|--------------|
| Module 1 | Color Science power distribution curve 15 0 | 3             | As specified |
| Module 2 | Digital Imaging 15 0                        | 7             | As specified |
| Module 3 | Halftone reproduction 15 0                  | 11            | As specified |
| Module 4 | Direction, Porosity, Smoothness. 15 0       | 12            | As specified |

## MODULE 1

# Color Science power distribution curve 15 0

This module is presented from the official syllabus scope for Fundamentals of Printing Science. Use the topic cards for learning and the source transcript for exact coverage.

**As specified**  
official hours

**CO-linked**  
see outcomes

### Learning goals

- Explain the official Module 1 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

### Key facts

- Official module title: Color Science power distribution curve 15 0
- Official duration: As specified
- Official topic entries captured: 3
- The full source excerpt is preserved below for coverage checking.

## perception and Color reproduction. Maxwell Color Triangle.

perception and Color reproduction. Maxwell Color Triangle. is an official topic in Module 1 of Fundamentals of Printing Science. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify perception and Color reproduction. Maxwell Color Triangle. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, perception and color reproduction. maxwell color triangle. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

#### Topic study guide

| Aspect              | What to prepare                                                                                      |
|---------------------|------------------------------------------------------------------------------------------------------|
| Definition/identity | What perception and Color reproduction. Maxwell Color Triangle. means in the official course context |
| Study lens          | Principle → components or stages → result → application                                              |
| Exam evidence       | Definition, labelled explanation, comparison, procedure or example as appropriate                    |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related

but different term.

**Malayalam support:** Module 1-പ്രകാശം perception and Color reproduction. Maxwell Color Triangle. പരസ്പരം definition/meaning പരസ്പരം. പരസ്പരം പരസ്പരം, steps, application പരസ്പരം പരസ്പരം പരസ്പരം. Technical terms English-പ്രകാശം പരസ്പരം പരസ്പരം.

## Scanners - Working Principle & Types.

Scanners - Working Principle & Types. is an official topic in Module 1 of Fundamentals of Printing Science. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify Scanners - Working Principle & Types. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, scanners - working principle & types. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

#### Topic study guide

| Aspect              | What to prepare                                                                   |
|---------------------|-----------------------------------------------------------------------------------|
| Definition/identity | What Scanners - Working Principle & Types. means in the official course context   |
| Study lens          | Principle → components or stages → result → application                           |
| Exam evidence       | Definition, labelled explanation, comparison, procedure or example as appropriate |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 1-Scanner Scanners - Working Principle & Types.

Scanner definition/meaning. Scanner types, steps, application. Technical terms English-Malayalam.

## Digital image - Pixel and bit depth. Vector file formats and raster file formats.

Digital image - Pixel and bit depth. Vector file formats and raster file formats. is an official topic in Module 1 of Fundamentals of Printing Science. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify Digital image - Pixel and bit depth. Vector file formats and raster file formats. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, digital image - pixel and bit depth. vector file formats and raster file formats. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

### Topic study guide

| Aspect              | What to prepare                                                                                                             |
|---------------------|-----------------------------------------------------------------------------------------------------------------------------|
| Definition/identity | What Digital image - Pixel and bit depth. Vector file formats and raster file formats. means in the official course context |
| Study lens          | Principle → components or stages → result → application                                                                     |
| Exam evidence       | Definition, labelled explanation, comparison, procedure or example as appropriate                                           |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related

but different term.

**Malayalam support:** Module 1- ഡിജിറ്റൽ ഫോട്ടോ ഫോർമാറ്റ്. വെക്റ്റർ ഫോർമാറ്റ് ഫയലുകൾ. ഡിജിറ്റൽ ഫോട്ടോ ഫോർമാറ്റ്. ഡിജിറ്റൽ ഫോട്ടോ ഫോർമാറ്റ്, steps, application ഡിജിറ്റൽ ഫോട്ടോ ഫോർമാറ്റ്. Technical terms English-ൽ ഉൾപ്പെടെ ഉൾപ്പെടെ.

## Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module. Educational explanations below are separate elaboration.

|                                                            |                                                                                     |
|------------------------------------------------------------|-------------------------------------------------------------------------------------|
| Color Science                                              | power distribution curve                                                            |
| 15                                                         | 0                                                                                   |
| Monochromatic, Analog and Complementary color, Metamerism. | Color Schemes: Warm color, Cool color, color. Color wheel. Memory                   |
| color theories. Color theories Vs Color Color Triangle.    | Color Theories: Additive and Subtractive perception and Color reproduction. Maxwell |
| color separation                                           | Conventional color separation & Electronic                                          |
| Vector file formats and raster file formats.               | Scanners - Working Principle & Types. Digital image - Pixel and bit depth.          |

**Module study routine:** Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

## MODULE 2

# Digital Imaging 15 0

This module is presented from the official syllabus scope for Fundamentals of Printing Science. Use the topic cards for learning and the source transcript for exact coverage.

**As specified**  
official hours

**CO-linked**  
see outcomes

## Learning goals

- Explain the official Module 2 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

## Key facts

- Official module title: Digital Imaging 15 0
- Official duration: As specified
- Official topic entries captured: 7
- The full source excerpt is preserved below for coverage checking.

## Digital Imaging 15 0

Digital Imaging 15 0 is an official topic in Module 2 of Fundamentals of Printing Science. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify Digital Imaging 15 0 using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, digital imaging 15 0 should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

#### Topic study guide

| Aspect              | What to prepare                                                                   |
|---------------------|-----------------------------------------------------------------------------------|
| Definition/identity | What Digital Imaging 15 0 means in the official course context                    |
| Study lens          | Principle → components or stages → result → application                           |
| Exam evidence       | Definition, labelled explanation, comparison, procedure or example as appropriate |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 2- ഡിജിറ്റൽ ഇമേജിംഗ് 15 0 ഡിജിറ്റൽ ഇമേജിംഗ് ന്റെ  
definition/meaning എന്താണ്. ഡിജിറ്റൽ ഇമേജിംഗ് ന്റെ steps, application  
എന്താണ്. Technical terms English-ൽ എഴുതേണ്ടതാണ്.

## Different File formats- their features, and uses in printing - JPEG/JPG, TIFF, PNG, GIF,

Different File formats- their features, and uses in printing - JPEG/JPG, TIFF, PNG, GIF, is an official topic in Module 2 of Fundamentals of Printing Science. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify Different File formats- their features, and uses in printing - JPEG/JPG, TIFF, PNG, GIF, using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, different file formats- their features, and uses in printing - jpeg/jpg, tiff, png, gif, should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

#### Topic study guide

| Aspect              | What to prepare                                                                                                                    |
|---------------------|------------------------------------------------------------------------------------------------------------------------------------|
| Definition/identity | What Different File formats- their features, and uses in printing - JPEG/JPG, TIFF, PNG, GIF, means in the official course context |
| Study lens          | Principle → components or stages → result → application                                                                            |
| Exam evidence       | Definition, labelled explanation, comparison, procedure or example as appropriate                                                  |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 2-പ്രകൃതി വ്യത്യാസം Different File formats- their features, and uses in printing - JPEG/JPG, TIFF, PNG, GIF, പ്രകൃതി വ്യത്യാസം definition/meaning പ്രകൃതി വ്യത്യാസം. പ്രകൃതി വ്യത്യാസം പ്രകൃതി വ്യത്യാസം, steps, application പ്രകൃതി വ്യത്യാസം പ്രകൃതി വ്യത്യാസം. Technical terms English-പ്രകൃതി വ്യത്യാസം പ്രകൃതി വ്യത്യാസം.

## **BMP, PDF, JDF, EPS, PSD, CRD, DOC/DOCX, PPT/PPTX, FLASH-SWF, ZIP, HTML, QXD,**

BMP, PDF, JDF, EPS, PSD, CRD, DOC/DOCX, PPT/PPTX, FLASH-SWF, ZIP, HTML, QXD, is an official topic in Module 2 of Fundamentals of Printing Science. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### **Concept and explanation**

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### **Study points**

- Define or identify BMP, PDF, JDF, EPS, PSD, CRD, DOC/DOCX, PPT/PPTX, FLASH-SWF, ZIP, HTML, QXD, using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### **Engineering or professional relevance**

In Polytechnic practice, bmp, pdf, jdf, eps, psd, crd, doc/docx, ppt/pptx, flash-swf, zip, html, qxd, should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

#### **Topic study guide**

| Aspect              | What to prepare                                                                                                        |
|---------------------|------------------------------------------------------------------------------------------------------------------------|
| Definition/identity | What BMP, PDF, JDF, EPS, PSD, CRD, DOC/DOCX, PPT/PPTX, FLASH-SWF, ZIP, HTML, QXD, means in the official course context |
| Study lens          | Principle → components or stages → result → application                                                                |
| Exam evidence       | Definition, labelled explanation, comparison, procedure or example as appropriate                                      |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related

but different term.

**Malayalam support:** Module 2- BMP, PDF, JDF, EPS, PSD, CRD, DOC/DOCX, PPT/PPTX, FLASH-SWF, ZIP, HTML, QXD, definition/meaning, steps, application, Technical terms English-

## PMD, INDD, SLA/SCD, TXT, ODT, ICC/ICM, XML.

PMD, INDD, SLA/SCD, TXT, ODT, ICC/ICM, XML. is an official topic in Module 2 of Fundamentals of Printing Science. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify PMD, INDD, SLA/SCD, TXT, ODT, ICC/ICM, XML. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, pmd, indd, sla/scd, txt, odt, icc/icm, xml. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

#### Topic study guide

| Aspect              | What to prepare                                                                       |
|---------------------|---------------------------------------------------------------------------------------|
| Definition/identity | What PMD, INDD, SLA/SCD, TXT, ODT, ICC/ICM, XML. means in the official course context |
| Study lens          | Principle → components or stages → result → application                               |
| Exam evidence       | Definition, labelled explanation, comparison, procedure or example as appropriate     |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 2-പിന്നെ PMD, INDD, SLA/SCD, TXT, ODT, ICC/ICM, XML. പദപരിഭാഷണം പദപരിഭാഷണം definition/meaning പദപരിഭാഷണം. പദപരിഭാഷണം പദപരിഭാഷണം, steps, application പദപരിഭാഷണം പദപരിഭാഷണം പദപരിഭാഷണം. Technical terms English-പിന്നെ പദപരിഭാഷണം പദപരിഭാഷണം.

## Type of Originals - Lineoriginal,Continuous tone originals, Halftone original,Areas of

Type of Originals - Lineoriginal,Continuous tone originals, Halftone original,Areas of is an official topic in Module 2 of Fundamentals of Printing Science. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify Type of Originals - Lineoriginal,Continuous tone originals, Halftone original,Areas of using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, type of originals - lineoriginal,continuous tone originals, halftone original,areas of should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

#### Topic study guide

| Aspect              | What to prepare                                                                                                                  |
|---------------------|----------------------------------------------------------------------------------------------------------------------------------|
| Definition/identity | What Type of Originals - Lineoriginal,Continuous tone originals, Halftone original,Areas of means in the official course context |
| Study lens          | Principle → components or stages → result → application                                                                          |
| Exam evidence       | Definition, labelled explanation, comparison, procedure or example as appropriate                                                |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 2-രേഖാവിധി Type of Originals - Lineoriginal, Continuous tone originals, Halftone original, Areas of രേഖാവിധി definition/meaning രേഖാവിധി. രേഖാവിധി രേഖാവിധി, steps, application രേഖാവിധി രേഖാവിധി രേഖാവിധി. Technical terms English-രേഖാവിധി രേഖാവിധി.

## Halftone screening - AM, FM & Hybrid Screening process.

Halftone screening - AM, FM & Hybrid Screening process. is an official topic in Module 2 of Fundamentals of Printing Science. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify Halftone screening - AM, FM & Hybrid Screening process. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, halftone screening - am, fm & hybrid screening process. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

#### Topic study guide

| Aspect              | What to prepare                                                                                   |
|---------------------|---------------------------------------------------------------------------------------------------|
| Definition/identity | What Halftone screening - AM, FM & Hybrid Screening process. means in the official course context |
| Study lens          | Principle → components or stages → result → application                                           |
| Exam evidence       | Definition, labelled explanation, comparison, procedure or example as appropriate                 |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related

but different term.

**Malayalam support:** Module 2-½ Halftone screening - AM, FM & Hybrid Screening process. ഓരോരും ഓരോരും definition/meaning ഓരോരും ഓരോരും ഓരോരും, steps, application ഓരോരും ഓരോരും ഓരോരും. Technical terms English-½ ഓരോരും ഓരോരും ഓരോരും.

## Halftone dots - features and types.

Halftone dots - features and types. is an official topic in Module 2 of Fundamentals of Printing Science. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify Halftone dots - features and types. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, halftone dots - features and types. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

#### Topic study guide

| Aspect              | What to prepare                                                                   |
|---------------------|-----------------------------------------------------------------------------------|
| Definition/identity | What Halftone dots - features and types. means in the official course context     |
| Study lens          | Principle → components or stages → result → application                           |
| Exam evidence       | Definition, labelled explanation, comparison, procedure or example as appropriate |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 2-½ Halftone dots - features and types.

പ്രാഥമികമായി ഈ മോഡ്യൂളിൽ definition/meaning, steps, application എന്നിവയെക്കുറിച്ചാണ് പഠിക്കുക. Technical terms English-ൽ പരിചയപ്പെടുക.

**Official syllabus detail and terminology**

The following excerpt preserves official syllabus terminology for this module. Educational explanations below are separate elaboration.

Digital Imaging  
15 0

Different File formats- their features, and uses in printing - JPEG/JPG, TIFF, PNG, GIF, BMP, PDF, JDF, EPS, PSD, CRD, DOC/DOCX, PPT/PPTX, FLASH-SWF, ZIP, HTML, QXD, PMD, INDD, SLA/SCD, TXT, ODT, ICC/ICM, XML.

Type of Originals - Lineoriginal, Continuous tone originals, Halftone original, Areas of continuous tone image.

Screening process. Halftone screening - AM, FM & Hybrid

Halftone dots - features and types.

**Module study routine:** Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

**MODULE 3**

# Halftone reproduction 15 0

This module is presented from the official syllabus scope for Fundamentals of Printing Science. Use the topic cards for learning and the source transcript for exact coverage.

**As specified**  
official hours

**CO-linked**  
see outcomes

## Learning goals

- Explain the official Module 3 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

## Key facts

- Official module title: Halftone reproduction 15 0
- Official duration: As specified
- Official topic entries captured: 11
- The full source excerpt is preserved below for coverage checking.

## Halftone reproduction 15 0

Halftone reproduction 15 0 is an official topic in Module 3 of Fundamentals of Printing Science. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify Halftone reproduction 15 0 using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, halftone reproduction 15 0 should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

#### Topic study guide

| Aspect              | What to prepare                                                                   |
|---------------------|-----------------------------------------------------------------------------------|
| Definition/identity | What Halftone reproduction 15 0 means in the official course context              |
| Study lens          | Principle → components or stages → result → application                           |
| Exam evidence       | Definition, labelled explanation, comparison, procedure or example as appropriate |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 3-½ Halftone reproduction 15 0 ഫലപ്രസാദം  
ഫലപ്രസാദം definition/meaning ഫലപ്രസാദം. ഫലപ്രസാദം ഫലപ്രസാദം, steps,  
application ഫലപ്രസാദം ഫലപ്രസാദം ഫലപ്രസാദം. Technical terms English-ഫലപ്രസാദം  
ഫലപ്രസാദം.

## Optical Density - Definition, plays a role in printing. Tonal Gradation - Definition.

Optical Density - Definition, plays a role in printing. Tonal Gradation - Definition. is an official topic in Module 3 of Fundamentals of Printing Science. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify Optical Density - Definition, plays a role in printing. Tonal Gradation - Definition. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, optical density - definition, plays a role in printing. tonal gradation - definition. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

#### Topic study guide

| Aspect              | What to prepare                                                                                                                 |
|---------------------|---------------------------------------------------------------------------------------------------------------------------------|
| Definition/identity | What Optical Density - Definition, plays a role in printing. Tonal Gradation - Definition. means in the official course context |
| Study lens          | Principle → components or stages → result → application                                                                         |
| Exam evidence       | Definition, labelled explanation, comparison, procedure or example as appropriate                                               |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 3-പ്രകാശ സാന്ദ്രത - Definition, plays a role in printing. Tonal Gradation - Definition. പ്രകാശ സാന്ദ്രത definition/meaning പ്രകാശ സാന്ദ്രത. പ്രകാശ സാന്ദ്രത പ്രകാശ സാന്ദ്രത, steps, application പ്രകാശ സാന്ദ്രത പ്രകാശ സാന്ദ്രത. Technical terms English-പ്രകാശ സാന്ദ്രത പ്രകാശ സാന്ദ്രത.

## Halftone image - Significance and basics of halftone imaging.

Halftone image - Significance and basics of halftone imaging. is an official topic in Module 3 of Fundamentals of Printing Science. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify Halftone image - Significance and basics of halftone imaging. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, halftone image - significance and basics of halftone imaging. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

#### Topic study guide

| Aspect              | What to prepare                                                                                         |
|---------------------|---------------------------------------------------------------------------------------------------------|
| Definition/identity | What Halftone image - Significance and basics of halftone imaging. means in the official course context |
| Study lens          | Principle → components or stages → result → application                                                 |
| Exam evidence       | Definition, labelled explanation, comparison, procedure or example as appropriate                       |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related

but different term.

**Malayalam support:** Module 3- Half tone image - Significance and basics of halftone imaging. Half tone definition/meaning. Half tone steps, application. Technical terms English- Half tone image.

## Light sensitive material - Film (positive and negative), basics of image formation on

Light sensitive material - Film (positive and negative), basics of image formation on is an official topic in Module 3 of Fundamentals of Printing Science. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify Light sensitive material - Film (positive and negative), basics of image formation on using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, light sensitive material - film (positive and negative), basics of image formation on should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

#### Topic study guide

| Aspect              | What to prepare                                                                                                                 |
|---------------------|---------------------------------------------------------------------------------------------------------------------------------|
| Definition/identity | What Light sensitive material - Film (positive and negative), basics of image formation on means in the official course context |
| Study lens          | Principle → components or stages → result → application                                                                         |
| Exam evidence       | Definition, labelled explanation, comparison, procedure or example as appropriate                                               |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 3-പ്രകാശ സെൻസിറ്റീവ് മെറ്റീരിയൽ - ഫിലിം (പോസിറ്റീവ് and നെഗറ്റീവ്), ഫോട്ടോഗ്രാഫിയിൽ ചിത്ര രൂപീകരണത്തിന്റെ അടിസ്ഥാനം definition/meaning ഉൾപ്പെടെ. ഫോട്ടോഗ്രാഫിയിൽ ഫിലിം ഉപയോഗിക്കുന്ന ഫോട്ടോ എക്സ്പോഷർ, steps, application ഉൾപ്പെടെ. Technical terms English-ൽ ഉൾപ്പെടെ.

## Paper and Board Properties: Define - Paper and Board- Sizes, Classifications and

Paper and Board Properties: Define - Paper and Board- Sizes, Classifications and is an official topic in Module 3 of Fundamentals of Printing Science. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify Paper and Board Properties: Define - Paper and Board- Sizes, Classifications and using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, paper and board properties: define - paper and board- sizes, classifications and should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

#### Topic study guide

| Aspect              | What to prepare                                                                                                            |
|---------------------|----------------------------------------------------------------------------------------------------------------------------|
| Definition/identity | What Paper and Board Properties: Define - Paper and Board- Sizes, Classifications and means in the official course context |
| Study lens          | Principle → components or stages → result → application                                                                    |
| Exam evidence       | Definition, labelled explanation, comparison, procedure or example as appropriate                                          |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 3-പേപ്പർ പേപ്പർ and Board Properties: Define - Paper and Board- Sizes, Classifications and പേപ്പർ പേപ്പർ definition/meaning പേപ്പർ പേപ്പർ. പേപ്പർ പേപ്പർ പേപ്പർ, steps, application പേപ്പർ പേപ്പർ പേപ്പർ. Technical terms English-പേപ്പർ പേപ്പർ പേപ്പർ.

## their uses

their uses is an official topic in Module 3 of Fundamentals of Printing Science. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify their uses using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, their uses should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

#### Topic study guide

| Aspect              | What to prepare                                                                   |
|---------------------|-----------------------------------------------------------------------------------|
| Definition/identity | What their uses means in the official course context                              |
| Study lens          | Principle → components or stages → result → application                           |
| Exam evidence       | Definition, labelled explanation, comparison, procedure or example as appropriate |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 3-  
definition/meaning . steps, application  
Technical terms English-

## Printability & Runnability Properties.

Printability & Runnability Properties. is an official topic in Module 3 of Fundamentals of Printing Science. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify Printability & Runnability Properties. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, printability & runnability properties. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

#### Topic study guide

| Aspect              | What to prepare                                                                   |
|---------------------|-----------------------------------------------------------------------------------|
| Definition/identity | What Printability & Runnability Properties. means in the official course context  |
| Study lens          | Principle → components or stages → result → application                           |
| Exam evidence       | Definition, labelled explanation, comparison, procedure or example as appropriate |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 3-പ്രിന്റബിലിറ്റി & റണ്ണബിലിറ്റി പ്രോപ്പർട്ടീസ്.

പ്രിന്റബിലിറ്റി & റണ്ണബിലിറ്റി പ്രോപ്പർട്ടീസ് definition/meaning പ്രിന്റബിലിറ്റി & റണ്ണബിലിറ്റി പ്രോപ്പർട്ടീസ്, steps, application പ്രിന്റബിലിറ്റി & റണ്ണബിലിറ്റി പ്രോപ്പർട്ടീസ്. Technical terms English-ൽ പ്രിന്റബിലിറ്റി & റണ്ണബിലിറ്റി പ്രോപ്പർട്ടീസ്.

## Optical properties- Color, Gloss, Opacity, Metamerism, Brightness, Whiteness.

Optical properties- Color, Gloss, Opacity, Metamerism, Brightness, Whiteness. is an official topic in Module 3 of Fundamentals of Printing Science. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify Optical properties- Color, Gloss, Opacity, Metamerism, Brightness, Whiteness. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, optical properties- color, gloss, opacity, metamerism, brightness, whiteness. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

#### Topic study guide

| Aspect              | What to prepare                                                                                                         |
|---------------------|-------------------------------------------------------------------------------------------------------------------------|
| Definition/identity | What Optical properties- Color, Gloss, Opacity, Metamerism, Brightness, Whiteness. means in the official course context |
| Study lens          | Principle → components or stages → result → application                                                                 |
| Exam evidence       | Definition, labelled explanation, comparison, procedure or example as appropriate                                       |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related

but different term.

**Malayalam support:** Module 3-പ്രകാശ ഗുണങ്ങൾ- Color, Gloss, Opacity, Metamerism, Brightness, Whiteness. പദപരിഭാഷണം definition/meaning പദപരിഭാഷണം. പദപരിഭാഷണം പദപരിഭാഷണം, steps, application പദപരിഭാഷണം പദപരിഭാഷണം. Technical terms English-പ്രകാശ ഗുണങ്ങൾ.

## Structural Properties- Density, Basis Weight and Grammage, Caliper and Bulk, Wire

Structural Properties- Density, Basis Weight and Grammage, Caliper and Bulk, Wire is an official topic in Module 3 of Fundamentals of Printing Science. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify Structural Properties- Density, Basis Weight and Grammage, Caliper and Bulk, Wire using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, structural properties- density, basis weight and grammage, caliper and bulk, wire should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

#### Topic study guide

| Aspect              | What to prepare                                                                                                             |
|---------------------|-----------------------------------------------------------------------------------------------------------------------------|
| Definition/identity | What Structural Properties- Density, Basis Weight and Grammage, Caliper and Bulk, Wire means in the official course context |
| Study lens          | Principle → components or stages → result → application                                                                     |
| Exam evidence       | Definition, labelled explanation, comparison, procedure or example as appropriate                                           |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 3- $\square\square$  Structural Properties- Density, Basis Weight and Grammage, Caliper and Bulk, Wire  $\square\square\square\square\square\square\square\square\square\square$   $\square\square\square\square$  definition/meaning  $\square\square\square\square\square\square\square\square\square\square$ .  $\square\square\square\square\square$   $\square\square\square\square\square$   $\square\square\square\square\square\square$ , steps, application  $\square\square\square\square\square$   $\square\square\square\square\square\square\square\square$   $\square\square\square\square\square$ . Technical terms English- $\square$   $\square\square\square\square$   $\square\square\square\square\square\square\square\square\square$ .

## and Felt sides, Moisture Content and Relative Humidity, Dimensional Stability, Grain

and Felt sides, Moisture Content and Relative Humidity, Dimensional Stability, Grain is an official topic in Module 3 of Fundamentals of Printing Science. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify and Felt sides, Moisture Content and Relative Humidity, Dimensional Stability, Grain using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, and felt sides, moisture content and relative humidity, dimensional stability, grain should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

#### Topic study guide

| Aspect              | What to prepare                                                                                                                |
|---------------------|--------------------------------------------------------------------------------------------------------------------------------|
| Definition/identity | What and Felt sides, Moisture Content and Relative Humidity, Dimensional Stability, Grain means in the official course context |
| Study lens          | Principle → components or stages → result → application                                                                        |
| Exam evidence       | Definition, labelled explanation, comparison, procedure or example as appropriate                                              |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 3-പദപരിചയം and Felt sides, Moisture Content and Relative Humidity, Dimensional Stability, Grain പദപരിചയം definition/meaning പദപരിചയം. പദപരിചയം പദപരിചയം, steps, application പദപരിചയം പദപരിചയം. Technical terms English-പദപരിചയം പദപരിചയം.

## Paper and Ink Science

Paper and Ink Science is an official topic in Module 3 of Fundamentals of Printing Science. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify Paper and Ink Science using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, paper and ink science should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

#### Topic study guide

| Aspect              | What to prepare                                                                   |
|---------------------|-----------------------------------------------------------------------------------|
| Definition/identity | What Paper and Ink Science means in the official course context                   |
| Study lens          | Principle → components or stages → result → application                           |
| Exam evidence       | Definition, labelled explanation, comparison, procedure or example as appropriate |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 3-പേപ്പർ and Ink Science പേപ്പർ definition/meaning പേപ്പർ. പേപ്പർ പേപ്പർ പേപ്പർ, steps, application പേപ്പർ പേപ്പർ പേപ്പർ. Technical terms English-പേപ്പർ പേപ്പർ.

**Official syllabus detail and terminology**

The following excerpt preserves official syllabus terminology for this module. Educational explanations below are separate elaboration.

Halftone reproduction  
15 0  
Optical Density - Definition, plays a role in printing. Tonal Gradation - Definition. Halftone image - Significance and basics of halftone imaging.  
Light sensitive material - Film (positive and negative), basics of image formation on film  
Paper and Board Properties: Define - Paper and Board- Sizes, Classifications and their uses  
Printability & Runnability Properties. Optical properties- Color, Gloss, Opacity, Metamerism, Brightness, Whiteness. Structural Properties- Density, Basis Weight and Grammage, Caliper and Bulk, Wire and Felt sides, Moisture Content and Relative Humidity, Dimensional Stability, Grain  
Paper and Ink Science

**Module study routine:** Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

**MODULE 4**  
**Direction, Porosity, Smoothness. 15 0**

This module is presented from the official syllabus scope for Fundamentals of Printing Science. Use the topic cards for learning and the source transcript for exact coverage.

**As specified**  
official hours

**CO-linked**  
see outcomes

## Learning goals

- Explain the official Module 4 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

## Key facts

- Official module title: Direction, Porosity, Smoothness. 15 0
- Official duration: As specified
- Official topic entries captured: 12
- The full source excerpt is preserved below for coverage checking.

## Direction, Porosity, Smoothness. 15 0

Direction, Porosity, Smoothness. 15 0 is an official topic in Module 4 of Fundamentals of Printing Science. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify Direction, Porosity, Smoothness. 15 0 using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, direction, porosity, smoothness. 15 0 should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

#### Topic study guide

| Aspect              | What to prepare                                                                   |
|---------------------|-----------------------------------------------------------------------------------|
| Definition/identity | What Direction, Porosity, Smoothness. 15 0 means in the official course context   |
| Study lens          | Principle → components or stages → result → application                           |
| Exam evidence       | Definition, labelled explanation, comparison, procedure or example as appropriate |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 4-□□ Direction, Porosity, Smoothness. 15 0

日本語から英語へ訳す definition/meaning 日本語から英語へ訳す。 日本語から英語へ訳す, steps, application 日本語から英語へ訳す 日本語から英語へ訳す。 Technical terms English-日本語 日本語から英語へ訳す。

## in Printing Technology

in Printing Technology is an official topic in Module 4 of Fundamentals of Printing Science. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify in Printing Technology using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, in printing technology should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

#### Topic study guide

| Aspect              | What to prepare                                                                   |
|---------------------|-----------------------------------------------------------------------------------|
| Definition/identity | What in Printing Technology means in the official course context                  |
| Study lens          | Principle → components or stages → result → application                           |
| Exam evidence       | Definition, labelled explanation, comparison, procedure or example as appropriate |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 4-എന്നിയിലുള്ള Printing Technology എന്ന വിഷയത്തിന്റെ definition/meaning നൽകുക. എന്നിവിടെ, steps, application എന്നിവയെക്കുറിച്ചും വിവരിക്കുക. Technical terms English-ൽ എഴുതുക.

## Mechanical properties- Tear Resistance, Tensile Strength, Bursting Strength, Folding

Mechanical properties- Tear Resistance, Tensile Strength, Bursting Strength, Folding is an official topic in Module 4 of Fundamentals of Printing Science. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify Mechanical properties- Tear Resistance, Tensile Strength, Bursting Strength, Folding using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, mechanical properties- tear resistance, tensile strength, bursting strength, folding should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

#### Topic study guide

| Aspect              | What to prepare                                                                                                                |
|---------------------|--------------------------------------------------------------------------------------------------------------------------------|
| Definition/identity | What Mechanical properties- Tear Resistance, Tensile Strength, Bursting Strength, Folding means in the official course context |
| Study lens          | Principle → components or stages → result → application                                                                        |
| Exam evidence       | Definition, labelled explanation, comparison, procedure or example as appropriate                                              |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 4-  
Mechanical properties- Tear Resistance, Tensile Strength, Bursting Strength, Folding   
 definition/meaning   
 .   
 , steps, application   
 . Technical terms English-  
 .

## Endurance, Stiffness, Abrasion Resistance.

Endurance, Stiffness, Abrasion Resistance. is an official topic in Module 4 of Fundamentals of Printing Science. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify Endurance, Stiffness, Abrasion Resistance. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, endurance, stiffness, abrasion resistance. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

#### Topic study guide

| Aspect              | What to prepare                                                                      |
|---------------------|--------------------------------------------------------------------------------------|
| Definition/identity | What Endurance, Stiffness, Abrasion Resistance. means in the official course context |
| Study lens          | Principle → components or stages → result → application                              |
| Exam evidence       | Definition, labelled explanation, comparison, procedure or example as appropriate    |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 4-എൻഡറൻസ്, സ്റ്റിഫ്നസ്സ്, അബ്രേഷൻ റെസിസ്റ്റൻസ്.

എൻഡറൻസ് എന്നതിന്റെ definition/meaning എന്താണ്. എൻഡറൻസ് ഉണ്ടാകാൻ പറ്റുന്ന steps, application എന്തെല്ലാം ആണ്. Technical terms English-ൽ എൻഡറൻസ് എങ്ങനെ എഴുതണം.

## **Ink properties: Optical properties - Color, Gloss, Opacity, Transparency.**

Ink properties: Optical properties - Color, Gloss, Opacity, Transparency. is an official topic in Module 4 of Fundamentals of Printing Science. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### **Concept and explanation**

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### **Study points**

- Define or identify Ink properties: Optical properties - Color, Gloss, Opacity, Transparency. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### **Engineering or professional relevance**

In Polytechnic practice, ink properties: optical properties - color, gloss, opacity, transparency. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

#### **Topic study guide**

| <b>Aspect</b>       | <b>What to prepare</b>                                                                                              |
|---------------------|---------------------------------------------------------------------------------------------------------------------|
| Definition/identity | What Ink properties: Optical properties - Color, Gloss, Opacity, Transparency. means in the official course context |
| Study lens          | Principle → components or stages → result → application                                                             |
| Exam evidence       | Definition, labelled explanation, comparison, procedure or example as appropriate                                   |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related

but different term.

**Malayalam support:** Module 4-ഇന്ക് പ്രകൃതി: പ്രകൃതി പ്രകൃതി - Color, Gloss, Opacity, Transparency. ഇന്ക് പ്രകൃതി definition/meaning ഇന്ക് പ്രകൃതി പ്രകൃതി. ഇന്ക് പ്രകൃതി പ്രകൃതി, steps, application ഇന്ക് പ്രകൃതി പ്രകൃതി. Technical terms English-ഇന്ക് പ്രകൃതി പ്രകൃതി.

## Physical properties - Length, Viscosity, Tack, Flow, Adhesion, Scuff resistance.

Physical properties - Length, Viscosity, Tack, Flow, Adhesion, Scuff resistance. is an official topic in Module 4 of Fundamentals of Printing Science. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify Physical properties - Length, Viscosity, Tack, Flow, Adhesion, Scuff resistance. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, physical properties - length, viscosity, tack, flow, adhesion, scuff resistance. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

#### Topic study guide

| Aspect              | What to prepare                                                                                                            |
|---------------------|----------------------------------------------------------------------------------------------------------------------------|
| Definition/identity | What Physical properties - Length, Viscosity, Tack, Flow, Adhesion, Scuff resistance. means in the official course context |
| Study lens          | Principle → components or stages → result → application                                                                    |
| Exam evidence       | Definition, labelled explanation, comparison, procedure or example as appropriate                                          |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related

but different term.

**Malayalam support:** Module 4-പ്രകൃതിഗുണങ്ങൾ - Length, Viscosity, Tack, Flow, Adhesion, Scuff resistance. പദപരിഭാഷണം definition/meaning പദപരിഭാഷണം. പദപരിഭാഷണം പദപരിഭാഷണം, steps, application പദപരിഭാഷണം പദപരിഭാഷണം. Technical terms English-പദപരിഭാഷണം പദപരിഭാഷണം.

## Rheological properties - plastic, pseudoplastic, dilatant and thixotropic substances,

Rheological properties - plastic, pseudoplastic, dilatant and thixotropic substances, is an official topic in Module 4 of Fundamentals of Printing Science. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify Rheological properties - plastic, pseudoplastic, dilatant and thixotropic substances, using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, rheological properties - plastic, pseudoplastic, dilatant and thixotropic substances, should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

#### Topic study guide

| Aspect              | What to prepare                                                                                                                 |
|---------------------|---------------------------------------------------------------------------------------------------------------------------------|
| Definition/identity | What Rheological properties - plastic, pseudoplastic, dilatant and thixotropic substances, means in the official course context |
| Study lens          | Principle → components or stages → result → application                                                                         |
| Exam evidence       | Definition, labelled explanation, comparison, procedure or example as appropriate                                               |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 4- രൂപകാന്ത ഗുണങ്ങൾ - പ്ലാസ്റ്റിക്, പ്സെഡോപ്ലാസ്റ്റിക്, ഡിലാറ്റന്റും തിക്സോട്രോപ്പിക് സബ്സ്റ്റാൻസുകളും, രൂപകാന്ത ഗുണങ്ങളുടെ നിർവ്വചന/അർത്ഥം, രൂപകാന്ത ഗുണങ്ങളുടെ ഘട്ടങ്ങൾ, പ്രയോഗ രൂപകാന്ത ഗുണങ്ങളുടെ പ്രയോഗങ്ങൾ. Technical terms English-ൽ രൂപകാന്ത ഗുണങ്ങളുടെ നിർവ്വചനം.

## viscoelasticity.

viscoelasticity. is an official topic in Module 4 of Fundamentals of Printing Science. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify viscoelasticity. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, viscoelasticity. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

#### Topic study guide

| Aspect              | What to prepare                                                                   |
|---------------------|-----------------------------------------------------------------------------------|
| Definition/identity | What viscoelasticity. means in the official course context                        |
| Study lens          | Principle → components or stages → result → application                           |
| Exam evidence       | Definition, labelled explanation, comparison, procedure or example as appropriate |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 4- $\square$  viscoelasticity.  $\square\square\square\square\square\square\square\square\square\square$  definition/meaning  $\square\square\square\square\square\square\square\square\square\square$ .  $\square\square\square\square\square$  steps, application  $\square\square\square\square\square$  Technical terms English- $\square$

## Detailed Syllabus

Detailed Syllabus is an official topic in Module 4 of Fundamentals of Printing Science. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify Detailed Syllabus using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, detailed syllabus should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

#### Topic study guide

| Aspect              | What to prepare                                                                   |
|---------------------|-----------------------------------------------------------------------------------|
| Definition/identity | What Detailed Syllabus means in the official course context                       |
| Study lens          | Principle → components or stages → result → application                           |
| Exam evidence       | Definition, labelled explanation, comparison, procedure or example as appropriate |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 4- Detailed Syllabus  $\frac{1}{2}$  definition/meaning  $\frac{1}{2}$ .  $\frac{1}{2}$   $\frac{1}{2}$   $\frac{1}{2}$ , steps, application  $\frac{1}{2}$   $\frac{1}{2}$   $\frac{1}{2}$ . Technical terms English-  $\frac{1}{2}$   $\frac{1}{2}$   $\frac{1}{2}$ .

## Mode of

Mode of is an official topic in Module 4 of Fundamentals of Printing Science. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify Mode of using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, mode of should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

#### Topic study guide

| Aspect              | What to prepare                                                                   |
|---------------------|-----------------------------------------------------------------------------------|
| Definition/identity | What Mode of means in the official course context                                 |
| Study lens          | Principle → components or stages → result → application                           |
| Exam evidence       | Definition, labelled explanation, comparison, procedure or example as appropriate |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 4-Mode of  $\frac{a+b}{2}$  definition/meaning  $\frac{a+b}{2}$ .  $\frac{a+b}{2}$   $\frac{a+b}{2}$ , steps, application  $\frac{a+b}{2}$   $\frac{a+b}{2}$ . Technical terms English- $\frac{a+b}{2}$   $\frac{a+b}{2}$ .

## Mapped Instructional

Mapped Instructional is an official topic in Module 4 of Fundamentals of Printing Science. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify Mapped Instructional using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, mapped instructional should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

#### Topic study guide

| Aspect              | What to prepare                                                                   |
|---------------------|-----------------------------------------------------------------------------------|
| Definition/identity | What Mapped Instructional means in the official course context                    |
| Study lens          | Principle → components or stages → result → application                           |
| Exam evidence       | Definition, labelled explanation, comparison, procedure or example as appropriate |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 4-എന്ന Mapped Instructional ഉള്ളടക്കത്തിന്റെ ഉപയോഗം definition/meaning ഉള്ളടക്കത്തിന്റെ ഉപയോഗം. ഉള്ളടക്കം ഉള്ളടക്കം ഉള്ളടക്കം, steps, application ഉള്ളടക്കം ഉള്ളടക്കം ഉള്ളടക്കം. Technical terms English-എന്ന ഉള്ളടക്കം ഉള്ളടക്കം ഉള്ളടക്കം.

## Subtopic Student Learning Outcome delivery Taxonomy Level

Subtopic Student Learning Outcome delivery Taxonomy Level is an official topic in Module 4 of Fundamentals of Printing Science. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify Subtopic Student Learning Outcome delivery Taxonomy Level using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, subtopic student learning outcome delivery taxonomy level should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

#### Topic study guide

| Aspect              | What to prepare                                                                                     |
|---------------------|-----------------------------------------------------------------------------------------------------|
| Definition/identity | What Subtopic Student Learning Outcome delivery Taxonomy Level means in the official course context |
| Study lens          | Principle → components or stages → result → application                                             |
| Exam evidence       | Definition, labelled explanation, comparison, procedure or example as appropriate                   |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related

but different term.

**Malayalam support:** Module 4- Subtopic Student Learning Outcome delivery Taxonomy Level definition/meaning . steps, application . Technical terms English- .

## Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module. Educational explanations below are separate elaboration.

Direction, Porosity, Smoothness.

15

0

in Printing Technology

Tensile Strength, Bursting Strength, Folding

Mechanical properties- Tear Resistance,

Endurance, Stiffness, Abrasion Resistance.

Ink properties: Optical properties -

Color, Gloss, Opacity, Transparency.

Physical properties - Length, Viscosity,

Tack, Flow, Adhesion, Scuff resistance.

Rheological properties - plastic,

pseudoplastic, dilatant and thixotropic substances, viscoelasticity.

Detailed Syllabus

Mode of

Mapped

Instructional

Subtopic

Student Learning Outcome

delivery

Taxonomy Level

C0

Hours

(L/T)

**Module study routine:** Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

## Concept and formula bank

### Answer structure

Definition → Principle → Components/steps → Application → Conclusion

Use this sequence for descriptive questions when the topic does not require a numerical formula.

### Practical record

Aim → Apparatus → Procedure → Observation → Result → Safety

Use this sequence for laboratory, workshop and field activities.

## Diagram and source-transcript library

### Module 1 source and topic cards

Jump to official terminology, topic explanations and study guidance.

### Module 2 source and topic cards

Jump to official terminology, topic explanations and study guidance.

### Module 3 source and topic cards

Jump to official terminology, topic explanations and study guidance.

### Module 4 source and topic cards

## OFFICIAL SELF-LEARNING

### Activities from the syllabus

Use the extracted official activity wording as the record of required self-learning work.

- for Self-Learning
- Week Self-Learning description Hours Module I
- Create small design samples demonstrating: Warm color scheme Cool color scheme Monochromatic scheme Analogous scheme Complementary scheme
- Prepare a brief note explaining color and its attributes (hue, saturation, lightness) with real-life examples. 6
- Compare color appearance under daylight, tungsten, and LED light sources and prepare a report. 6 Module II
- Prepare a table listing different file formats used in printing and group formats as image, document, layout, color
- management, and workflow files. 12
- Mention their features, advantages, limitations, and typical printing applications.
- Explain the role of pixel count and bit depth in print quality. 12
- Draw a simple block diagram showing the working of a scanner. 12 Module III
- Take a photograph or printed picture and mark highlight, midtone, and shadow areas using a pencil or digital tool. Write a
- short note explaining how tonal variation appears in different areas.
- Identify and explain which screening method is most suitable for different print applications such as newspapers, packaging,
- and high-quality art prints. Module IV
- Prepare a one-page report on “Role of Paper and Ink Properties in Print Quality” with examples. 6
- Match paper types with suitable printing applications (newspaper, books, packaging, labels). 6
- Total Self-Learning hours 90 Learning Resources TEXT BOOK
- Title Author Publication
- Chemistry in Printing, 2nd Edition, 2011 Tulika Das Barnana Prakashani
- Understanding Digital Colour, 1995 P. Green GATF REFERENCE

## EXAMINATION PREPARATION

# Assessment methodology

The following source extract preserves the assessment information detected in the official PDF. Verify mark conversions and institutional updates against the current source before an examination.

|                                                                                                                                                                                                             |                                                                                                                           |                                                                                                                                |                                                                                                              |                                               |     |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------|-----------------------------------------------|-----|
| – Theory                                                                                                                                                                                                    |                                                                                                                           |                                                                                                                                |                                                                                                              |                                               |     |
| CA3                                                                                                                                                                                                         | CA4                                                                                                                       | CA1<br>CA5                                                                                                                     | CA6                                                                                                          | CA2                                           | ESE |
| Mode<br>Self-learning<br>activities<br>(Module 3)<br>Duration<br>2 hours<br>Exam mark<br>15<br>Converted to<br>20<br>Marks<br>20<br>Total<br>60<br>Tentative<br>End of<br>11th week<br>schedule<br>semester | Attendance<br>Self-learning<br>activities<br>(Module 4)<br>2 hours<br>15<br>5<br>60<br>End of<br>By 14th week<br>semester | Self-learning<br>Series<br>activities<br>Exam<br>(Module 1)<br>(2 modules)<br>2.5 hours<br>15<br>50<br>By 4th week<br>8th week | Self-learning<br>Series<br>activities<br>Exam<br>(Module 2)<br>(2 modules)<br>50<br>By 7th week<br>15th week | Written<br>Exam<br>(theory)<br>15<br>40<br>By |     |

**Practice-paper notice:** The model paper below is generated for revision and is not an official examination paper.

## ACTIVE RECALL

### Module-wise questions and answers

These are practice questions based on official topic coverage, not official examination questions.

#### Module 1

Define or explain perception and Color reproduction. Maxwell Color Triangle..

**(3 marks)**

**Answer:** Begin with the standard definition or identity of *perception and Color reproduction. Maxwell Color Triangle*.. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

**Malayalam support:** □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

**Define or explain Scanners - Working Principle & Types.. (3 marks)**

**Answer:** Begin with the standard definition or identity of *Scanners - Working Principle & Types*.. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

**Malayalam support:** □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

**Define or explain Digital image - Pixel and bit depth. Vector file formats and raster file formats.. (3 marks)**

**Answer:** Begin with the standard definition or identity of *Digital image - Pixel and bit depth. Vector file formats and raster file formats*.. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

**Malayalam support:** □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

## Module 2

**Define or explain Digital Imaging 15 0. (3 marks)**

**Answer:** Begin with the standard definition or identity of *Digital Imaging 15 0*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

**Malayalam support:** □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

**Define or explain Different File formats- their features, and uses in printing - JPEG/JPG, TIFF, PNG, GIF,. (3 marks)**

**Answer:** Begin with the standard definition or identity of *Different File formats- their features, and uses in printing - JPEG/JPG, TIFF, PNG, GIF,.* State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

**Malayalam support:** □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

**Define or explain BMP, PDF, JDF, EPS, PSD, CRD, DOC/DOCX, PPT/PPTX, FLASH-SWF, ZIP, HTML, QXD,. (3 marks)**

**Answer:** Begin with the standard definition or identity of *BMP, PDF, JDF, EPS, PSD, CRD, DOC/DOCX, PPT/PPTX, FLASH-SWF, ZIP, HTML, QXD,.* State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

**Malayalam support:** □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

**Define or explain PMD, INDD, SLA/SCD, TXT, ODT, ICC/ICM, XML.. (3 marks)**

**Answer:** Begin with the standard definition or identity of *PMD, INDD, SLA/SCD, TXT, ODT, ICC/ICM, XML..* State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

**Malayalam support:** □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

## Module 3

### Define or explain Halftone reproduction 15 0. (3 marks)

**Answer:** Begin with the standard definition or identity of *Halftone reproduction 15 0*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

**Malayalam support:** □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

### Define or explain Optical Density - Definition, plays a role in printing. Tonal Gradation - Definition.. (3 marks)

**Answer:** Begin with the standard definition or identity of *Optical Density - Definition, plays a role in printing. Tonal Gradation - Definition..* State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

**Malayalam support:** □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

### Define or explain Halftone image - Significance and basics of halftone imaging.. (3 marks)

**Answer:** Begin with the standard definition or identity of *Halftone image - Significance and basics of halftone imaging..* State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

**Malayalam support:** □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

**Define or explain Light sensitive material - Film (positive and negative), basics of image formation on. (3 marks)**

**Answer:** Begin with the standard definition or identity of *Light sensitive material - Film (positive and negative), basics of image formation on*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

**Malayalam support:** □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

## Module 4

**Define or explain Direction, Porosity, Smoothness. 15 0. (3 marks)**

**Answer:** Begin with the standard definition or identity of *Direction, Porosity, Smoothness. 15 0*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

**Malayalam support:** □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

**Define or explain in Printing Technology. (3 marks)**

**Answer:** Begin with the standard definition or identity of *in Printing Technology*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

**Malayalam support:** □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

**Define or explain Mechanical properties- Tear Resistance, Tensile Strength, Bursting Strength, Folding. (3 marks)**

**Answer:** Begin with the standard definition or identity of *Mechanical properties- Tear Resistance, Tensile Strength, Bursting Strength, Folding*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

**Malayalam support:** □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

**Define or explain Endurance, Stiffness, Abrasion Resistance.. (3 marks)**

**Answer:** Begin with the standard definition or identity of *Endurance, Stiffness, Abrasion Resistance..* State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

**Malayalam support:** □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

## PRACTICE ONLY

# Practice Model Paper — Not an Official Examination Paper

Attempt one short definition, one structured explanation and one long answer from every module. Use the official assessment pattern reproduced above when selecting time and marks.

## Module 1

1-mark definition: State the meaning of **perception and Color reproduction. Maxwell Color Triangle..**

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

## Module 2

1-mark definition: State the meaning of **Digital Imaging 15 0**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

## Module 3

1-mark definition: State the meaning of **Halftone reproduction 15 0**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

## Module 4

1-mark definition: State the meaning of **Direction, Porosity, Smoothness. 15 0**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

### QUICK REVISION

## Exam checklist

### Before writing

- Read the command word: define, explain, compare, draw, calculate, analyse or design.

- Use the exact course terminology from the source transcript.
- Plan labels, units, sequence and marks before writing.

### Before submitting

- Check every module has been revised.
- Check diagrams, tables, formula symbols and units.
- Separate official syllabus information from personal elaboration.

## REFERENCE VOCABULARY

### Glossary

#### Study glossary

| Term              | Meaning in this lesson                                                   |
|-------------------|--------------------------------------------------------------------------|
| Course Outcome    | A capability the student should demonstrate after completing the course. |
| Module            | An official syllabus unit with defined topics and hours.                 |
| CIE               | Continuous Internal Evaluation.                                          |
| ESE               | End Semester Examination.                                                |
| Source transcript | A preserved extract of the official syllabus used for coverage checking. |

## SOURCE-SUPPORTED REFERENCES

### References and web resources

#### References listed in the official PDF

References are included in the official source PDF.

#### Web resources listed in the official PDF

- URL Modules Covered

## IN-PAGE SEARCH

## Search this lesson

Prepared from the supplied official SITTTR syllabus PDF for Course 2101. Verify institutional updates against the current official curriculum.